



Vojtěch Špaček

Senior C++ engineer — safety-critical embedded & HIL test automation | Linux/networking | agentic-AI (MCP) tooling

Pragmatic programmer and creative problem solver — systems, verification and open source.

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Profile Summary

Senior C++ engineer working on safety-critical and real-time systems: aircraft onboard diagnostics at Honeywell, electron-microscopy detector software at Thermo Fisher Scientific, and a Linux inspection proxy and firewall at Trusted Network Solutions.

Most recently sole owner of the hardware-in-the-loop verification suite and automated bench for a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), raising measured conformance to 259 of 271 tracked state transitions (95.6%).

Linux, networking and security specialist — Netlink, nftables, TLS inspection, Kerberos/LDAP/OCSP — with C++20 daemons shipped as Debian packages and a Rust reimplementations published on crates.io.

Builds agentic-AI tooling on the Model Context Protocol: an agent harness adopted into a daily client delivery workflow, and MCP-Prompts, published to npm, Docker Hub and the MCP directories (Glama, MCP Market, PulseMCP) with 97 stars and 21 forks.

Preferred Roles

- Test Automation / Verification Engineer (HIL, safety-critical) — Hardware-in-the-loop benches, requirement traceability, instrument control and test evidence reporting.
- Senior C++ Engineer — embedded, RTOS and real-time systems — State-machine engines, hardware abstraction layers, inter-process communication, high availability.
- Linux Systems, Networking and Security Engineer — TCP/IP, Netlink, nftables, encrypted traffic inspection, daemon design and packaging.
- AI/LLM Integration Engineer — MCP servers and agentic tooling — Model Context Protocol servers, prompt and agent workflow design, LLM provider APIs, CI/CD for AI-assisted delivery.

Work Experience

Test Automation Engineer (Python, C#)

Resideo (Ademco) (via CoolPeople (contractor))

Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), as sole owner of the verification test suite and of the automated bench that drives it. Owned delivery end to end: requirement traceability, issue tracking, branch and pull-request workflow, Conan packaging, CI integration and monthly client reporting.

- Designed an Arrange/Act/Assert validation pipeline in Python with eight independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.
- Built a data-driven generator expanding a YAML catalogue of 270+ state transitions, 38 states and 33 events into roughly 290 executable tests, removing hand-written duplication.
- Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering a pyvisa/USBTMC/SCPI programmable AC source, CAN serial-bus monitoring and Modbus relay and probe I/O.
- Implemented a correlated JSONL timeline placing every instrument command, signal edge and bus message on one axis, enabling root-cause analysis across three independent devices.
- Diagnosed and fixed a leaked USB handle that had been misreported as a hardware fault for months, restoring an entire blocked test family.
- Raised measured conformance to 259 of 271 tracked transitions (95.6%) and closed a ~148-case gap in which stimulus windows were never actually measured, bringing it to zero.
- Delivered an unattended nightly bench pipeline with per-key locking, crash and abort recovery, result journalling, Excel evidence workbooks, bilingual HTML dashboards and Mermaid failure graphs, cutting report generation from 15.7 s to 1.6 s.
- Investigated 25+ firmware and harness defects against the C/C++ firmware sources, shipping a standalone reproduction script for each.
- Migrated a C#/.NET WinForms serial-bus monitor into version control, fixed a log-parser crash and reduced its build and deployment to one command.
- Created an agentic AI harness of specialised agents, skills, prompt workflows, policy hooks and Model Context Protocol servers, adopted as a daily part of the delivery workflow.

New skills learned: Hardware-in-the-Loop Test Design, Safety-Certified Product Verification (UL/CSA, ANSI Z21.20, NFPA 54), Ignition State-Machine Domain Knowledge, pyvisa / USBTMC / SCPI Instrument Control, Modbus I/O Cards, CAN Serial-Bus Diagnostics, Deterministic Multi-Threaded Device Ownership, Test-Evidence & Traceability Reporting (openpyxl, HTML dashboards, Mermaid), Model Context Protocol Server Development, Agentic Workflow Design & Governance

Software Developer (C++)

Thermo Fisher Scientific (via Ixperta (contractor)) · 2023 – 2024

Electron microscopy software for high-resolution detector hardware operation and image acquisition. Built server-side C++ applications, FPGA-based data processing pipelines with gRPC networking, and backend logic for a C#/.NET WPF image processing and display application. Worked in an international Agile team with bi-weekly stakeholder demos.

- Implemented server-side software in C++, including hardware abstraction layers (HAL) and inter-process communication (IPC) using COM and MIDL on Windows.
- Developed embedded software for FPGA hardware, using gRPC network communication in C++, Python and Linux environments.
- Contributed to WPF application development on the backend using C# and COM/MIDL IPC with the HAL, and ensured graceful degradation of the application.
- Optimized the data acquisition pipeline and implemented robust error handling.
- Developed vscode-automation, an enterprise-grade VSCode extension integrating TICS code quality analysis with the OpenAI API for automated violation detection and AI-powered remediation, using TypeScript, the VSCode Extension API and Git API integration.
- Led 3 knowledge sharing sessions on GitHub Copilot best practices and presented bi-weekly stakeholder demos, including the final presentation of the vscode-automation innovation sprint.

New skills learned: Image Acquisition & Processing, C#, .NET, Component Object Model, Software Development for Windows, gRPC, FPGA development, VSCode Extension Development, TypeScript, OpenAI API Integration, Technical Presentation & Stakeholder Communication

Software Engineer II

Honeywell · 2022 – 2023

Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight. Architected and implemented the DiagTest framework, a state machine-based diagnostic test engine in C++ managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

- Designed a job-based execution system with DiagTestJob and DiagTestMsgQueueProc for asynchronous test orchestration, implementing message queue processing patterns for reliable real-time diagnostics.
- Engineered a test interference management system preventing conflicting diagnostic tests from running simultaneously, ensuring aircraft system safety and data integrity.
- Implemented four diagnostic test modes (data collection, passive monitoring, interactive, non-interactive) with timeout management, precondition handling and aviation-standard command/response protocols.
- Developed comprehensive integration tests in Python to validate communication protocols, state transitions and system stability under various failure scenarios.
- Built a real-time parameter monitoring system (MonitorManager) tracking aircraft system values during test execution with maintenance message correlation.
- Optimized data stream processing with multi-threading and timeout management, and ensured aviation safety standards compliance.

New skills learned: Embedded Software, ARM, RTOS, Clang, Cross-platform Compilation, Flatbuffers (binary data serialization, C++ headers generation), Conan (dependency management), Enterprise Architect (UML diagrams), Aviation Safety Standards, State Machine Design Patterns, TeamCity, GitLab CI/CD

Application Developer

Trusted Network Solutions · 2020 – 2022

Kernun Adaptive Firewall. Implemented a forward proxy in C++ on Linux featuring HTTP/HTTPS proxying with TLS/SSL inspection, OCSP integration for certificate validation, Kerberos authentication and LDAP user/group lookup, plus the ipmon network monitoring daemon driving nftables firewall rules from real-time IPv4/IPv6 address changes.

- Built comprehensive proxy infrastructure with certificate generation and management (CA, client certificates), a dynamic configuration builder/parser system, and a manager/partition architecture for modular daemon design.
- Integrated security components including Suricata (intrusion detection) and nftables (static firewall), aggregating logs and managing connection information in PostgreSQL.
- Developed DNS resolution and caching mechanisms, domain categorization and client authentication methods.
- Implemented real-time network interface monitoring in ipmon using NETLINK sockets, handling both IPv4 and IPv6 address management, with Unix domain sockets for IPC and dynamic nftables rule management.
- Implemented thread-safe, event-driven networking using modern C++ (mutexes, RAII, atomic file operations) with JSON-based configuration management and configurable delay mechanisms.
- Developed reporter_log, a C++ replacement of a Perl command, with PostgreSQL integration, structured log parsing and processing, Perl interoperability for legacy compatibility and database schema management.
- Built a configuration generation system in TypeScript/JavaScript with YAML-based high-level configuration processing, certificate generation automation and integration with the web backend for dynamic proxy configuration.
- Developed comprehensive testing infrastructure with a Perl test framework covering HTTP/HTTPS protocol testing, TLS/SSL inspection, Kerberos/LDAP authentication and OCSP validation.

New skills learned: TCP Proxy, HTTP Proxy, Encrypted Traffic Inspection, TLS 1.3, C++20, C++17, Boost Libraries, JSON-CPP, Kerberos/Heimdal, OCSP, LDAP, SQLite, Perl, JavaScript/TypeScript, YAML, Jenkins, Linux Networking, Netlink, UNIX Sockets, SSH Tunneling, Port Forwarding, VirtualBox, GDB

Independent IT Contractor

Self-employed

Independent contracting for business and consumer clients: CI/CD automation pipelines and Linux server administration for B2B customers, and IT support, consumer hardware retail and security consulting for B2C customers.

- B2B: CI/CD automation pipelines and Linux server administration.
- B2C: IT support freelancing, consumer hardware retail and security consulting.

New skills learned: Kali Linux, Wireshark, TeamCity, Docker, Self-studied the Certified Ethical Hacker syllabus (exam not taken)

Automation Developer & IT Systems Specialist

IBM · 2014 – 2016

Enterprise IT operations: infrastructure monitoring, incident management and an automation solution that replaced manual ticketing and server administration work.

- Developed an automation application that replaced manual workload in ticketing operations and basic server administration, using Bash, PowerShell and JavaScript for cross-platform server management and applying automata theory to design state-driven workflows.
- Monitored Unix/Windows servers and resolved production incidents, working hands-on with Active Directory and VMware virtualization.

New skills learned: RedHat Linux, Windows Server, Active Directory, VMware, Bash, PowerShell, JavaScript, Practical usage of automata theory

Skills

Embedded & RTOS

C++ (C++17/C++20) (Senior) 
Embedded software / ARM / RTOS (Medior) 
State machine design patterns (Senior) 
Hardware abstraction layers & IPC (COM/MIDL, Unix sockets) (Medior) 
FPGA-adjacent development & gRPC (Medior) 
Flatbuffers (binary serialization) (Medior) 
Qt/QML (Working knowledge) 

Networking & Security

Linux (Senior) 
Linux networking, TCP/IP, Netlink, nftables (Senior) 
Encrypted traffic inspection (TLS 1.3, OSCP) (Medior) 
Kerberos/Heimdal, LDAP, PostgreSQL, Suricata IDS (Medior) 
Forward/TCP/HTTP proxy design (Senior) 
Rust (Working knowledge) 

Tooling

Git (Senior) 
Bash (Senior) 
Docker / Podman (Medior) 
Conan (C/C++ dependency management) (Medior) 
CI/CD (Jenkins, TeamCity, GitLab CI, GitHub Actions) (Medior) 
PowerShell (Medior) 
Enterprise Architect (UML) (Working knowledge) 
GDB, Wireshark, VirtualBox (Medior) 

Verification & HIL

Python (Medior) 
Hardware-in-the-loop test design (Senior) 
pyvisa / USBTMC / SCPI instrument control (Medior) 
Modbus I/O cards & CAN serial-bus diagnostics (Medior) 
Safety-certified product verification (UL/CSA, ANSI Z21.20, NFPA 54) (Medior) 
Requirement traceability & test evidence reporting (openpyxl, HTML, Mermaid) (Medior) 
C# / .NET (Medior) 
Perl test frameworks (Medior) 

AI / LLM / MCP

Model Context Protocol server development (Senior) 
Agentic workflow design & governance (agents, skills, policy hooks) (Medior) 
Prompt engineering (RAG, few-shot, prompt caching, structured outputs, tool calling) (Medior) 
LLM provider APIs & SDK integration (Medior) 
JavaScript / TypeScript (Medior) 
Node.js (Medior) 

Open Source & Projects

MCP-Prompts — 97 stars, 21 forks

github.com/sparesparrow/mcp-prompts

Model Context Protocol server for managing prompt templates and LLM interactions, with four core MCP implementations (filesystem operations, memory management, AWS integration and generic MCP configurations). Implemented a hexagonal architecture and integration tests; ships Docker images across multiple server types, AWS deployment examples (S3, Lambda, DynamoDB, CloudWatch, IAM, ECR), vector-based semantic search with pgvector and real-time collaboration over Server-Sent Events. Published to npm and Docker Hub and listed across the MCP server directories.

TypeScript · Node.js · PostgreSQL + pgvector · Docker · REST + SSE · AWS

rust-network-mgr

github.com/sparesparrow/rust-network-mgr

Redesign and reimplementation of ipmon as a Rust daemon orchestrating dynamic firewall rules via Netlink with a minimal memory footprint. Event-driven architecture for real-time network state changes, with atomic firewall updates that prevent race conditions and inconsistent rulesets during topology changes, lock-free data structures and epoll-based multiplexing for high-throughput event processing.

Rust · Netlink · nftables · epoll · Lock-free data structures

rtp-midi — 2 stars

github.com/sparesparrow/rtp-midi

Modular Rust architecture for real-time MIDI, audio and LED control. Low-latency MIDI routing connects music production hardware to DAWs and addressable LED systems over network protocols. Includes an Android hub (Kotlin + Rust NDK) with a foreground service, AMidi NDK and mDNS discovery, plus an ESP32 visualizer using FastLED and dual-core FreeRTOS, in a dual-protocol design: RTP-MIDI for DAWs, OSC for LED devices.

Rust (66.5%) · RTP-MIDI · OSC · Android NDK (Kotlin) · ESP32 / FreeRTOS · Docker

ipmon

github.com/sparesparrow/ipmon

Netlink socket monitoring daemon that detects real-time IPv4/IPv6 address changes and updates nftables firewall rules accordingly. Built with modern C++20 using RAI, move semantics, thread-safe and atomic operations, an event-driven architecture, Unix domain sockets for IPC and JSON configuration management. Packaged for Debian-based systems with complete systemd hardening profiles for production deployment.

C++20 · Linux kernel / NETLINK · nftables · Multi-threading · systemd · Debian packaging

mcp-project-orchestrator

github.com/sparesparrow/mcp-project-orchestrator

Template orchestration with pattern recognition and code generation. JSON-driven orchestration analyses user input to identify design patterns and create initial project files from templates, with project/component template management, variable substitution, prompt template rendering and Mermaid diagram generation (flowcharts, sequence and class diagrams) rendered to SVG/PNG.

TypeScript · LLM integration · Pattern matching · Vector search · Mermaid

podman-desktop-extension-mcp

github.com/sparesparrow/podman-desktop-extension-mcp

Podman Desktop extension for secure MCP server management in containerized environments, combining multiple agent containers for collaboration and routing between them. Supports rootless runtimes, encrypted credential storage and real-time health monitoring.

TypeScript · Svelte · Podman · Container security

mcp-router

github.com/sparesparrow/mcp-router

Workflow orchestration system for Model Context Protocol services, with a drag-and-drop visual designer for building AI pipelines, dynamic service discovery and resilient routing for fault-tolerant workflows.

TypeScript · Node.js · React · Docker · Microservices

mcp-servers

github.com/sparesparrow/mcp-servers

A suite of Model Context Protocol servers: mermaid (diagram generator), solid (SOLID code review), prompt-manager (prompt orchestration and templates) and more, with visualization, persistence and prompt management tools.

TypeScript · Node.js · Docker · LLM integration · Microservices

hard-coder

github.com/sparesparrow/hard-coder

Early adoption of agentic design patterns and multi-step reasoning in AI code generation: a reasoning-engine assistant with explicit reasoning and planning capability, designed when LLMs lacked advanced reasoning without agentic frameworks. Transforms ideas into working code including tests and documentation. Evolution path: OpenAI GPT Store → Anthropic migration → CrewAI refactoring → Cursor IDE Composer Notepad.

Python · AI agent framework · Reasoning engine · Code generation

github-events

github.com/sparesparrow/github-events

Python service monitoring 23+ GitHub event types with analytics: repository health scoring, developer productivity analysis and security anomaly detection. FastAPI REST API with 15+ endpoints, an interactive dashboard with live data, MCP server support and dual backends (SQLite and AWS DynamoDB) with ecosystem monitoring for cross-domain cooperation.

Python · FastAPI · SQLite + DynamoDB · MCP server · Analytics dashboard

human-action

github.com/sparesparrow/human-action

Audiobook generation pipeline built on the ElevenLabs Python SDK, producing a Czech TTS narration of Ludwig von Mises' Human Action. Processes dense academic text into high-quality audio narration while maintaining natural intonation and pronunciation.

Python · FastAPI · ElevenLabs API · Voice synthesis · Multilingual processing

sparetools

github.com/sparesparrow/sparetools

Monorepo aggregating shared Python tools and Conan recipes, demonstrating C/C++ dependency management, modern Python packaging and cross-platform CI/CD.

Python · C/C++ · Conan · Cross-platform CI/CD

cpy

github.com/sparesparrow/cpy

Bundled CPython runtime for sparetools.

Python · CPython packaging

cursor-rules

github.com/sparesparrow/cursor-rules

Rules engine for the Cursor IDE that structures AI interactions, using glob-based file matching and hierarchical organization for consistent workflows.

Markdown · MDC format · LLM integration · Pattern matching

cursor-audio-notifications

github.com/sparesparrow/cursor-audio-notifications

Cursor IDE extension adding audio notifications for events such as build completion, improving workflow awareness and reducing context switching.

TypeScript · ElevenLabs audio synthesis · IDE integration

openssl-crypto-policies-test

github.com/sparesparrow/openssl-crypto-policies-test

Test suite validating OpenSSL cryptographic policy enforcement, with automated TLS security level verification and cipher benchmarking.

Bash · OpenSSL · Cryptography

btc-pay-server

github.com/sparesparrow/btc-pay-server

Self-hosted Bitcoin payment processor in Rust, offering a REST API for handling payments, generating payment requests and verifying transactions for easy integration into web applications.

Rust · Bitcoin protocol · REST API

Education & Training

Masaryk University, Faculty of Informatics — Computer Science

Coursework only — no academic title — Coursework in programming, Linux and networking.

Coursera.org / DeepLearning.AI — Machine learning and LLM engineering

Coursework only — no academic title — Deep Learning Specialization (four machine learning courses by Andrew Ng), plus data analysis, prompt engineering and building LLM agents.

Self-directed study — Design patterns, software architecture and offensive security

Coursework only — no academic title — Design Patterns (refactoring.guru) and Hack The Box cybersecurity training; self-studied the Certified Ethical Hacker syllabus (exam not taken).

Professional development reading — Software engineering practice

Coursework only — no academic title — Software Engineering at Google, The Pragmatic Programmer, Clean Architecture.

Languages

Czech (native) · Slovak (native) · English (C1) · German (A1)

Availability & Work Authorization

Available as an independent contractor (Czech IČO). Brno on-site and remote. Trading on a Czech IČO. EU citizen (Czech Republic) — no permit required in the EU/EEA.

Interests

Ludwig von Mises, Murray Rothbard · Ray Kurzweil, Nick Bostrom, Max Tegmark · Lex Fridman, Andrew Huberman, Dario Amodei · Neuroscience & Brain-Machine Interfaces · Music production, and producing a Czech audiobook — a TTS narration of Mises' Human Action

Online version: <https://sparesparrow.github.io/cv/> · © Vojtěch Špaček. All rights reserved.